

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: fos, jun (insert sizes are 4.2 and 1.8 kb using different vectors), linear
Species Used	Mouse, WEHI-3B, myelomonocytic leukemia		

Before the Pulse

Cell Growth Medium	McCoy's 5A (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Exponential
		Pre-pulse Incubation	10 min at room temperature

Wash Solution Phosphate Buffered Saline

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature	Cuvette Gap	0.4 cm
Electroporation Medium*	Phosphate Buffered Saline	Voltage	0.25 kV
Cell Density	1 x 10 ⁷ (7) cells / ml	Field Strength	0.625 kV/cm
Volume of Cells	0.8 ml	Capacitor	500 µF
DNA Concentration	15 µg /10 µg	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Phosphate Buffered Saline	Time Constant	14 msec
Volume of DNA	10 µl		
After the Pulse			
Outgrowth Medium	McCoy's 5A		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	2 weeks
Selection Method or Assay Used	G-418
Electroporation Efficiency	Not done
Per Cent Survival	50 to 75%

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Survey Number

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